

Math 150 – Exam 2 – Spring 2016

I pledge to demonstrate personal and academic integrity in all matters with my fellow students, the faculty, and the administration here at USD. I promise to be honest and accountable for my actions; and to uphold the stature of scholastic honesty to better myself and those around me. Initials: _____

Name: Solutions Section: **1 2**

Instructions: Read all directions carefully. Ask for clarification if you do not understand the directions. Show your work when necessary. Complete sentences and justifications are only necessary when indicated. There are 6 pages to this exam.

There are 100 points on this test and 4 Extra Credit points.

Num.	Score	Out of
1		5
2		12
3		16
4		5
5		20
6		30
7		12
B		*4*
Total		100

10:19 - 10:35

1. {4 points} Check in on your mindset. Write down a positive statement about your capacity to learn calculus.

with Time and hard work I can learn calculus

2. {3 points each} State whether each statement is True or False.

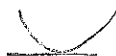
(a) If $y = e^{-x}$, then $y' = e^x$.

False

(b) $\frac{d}{dx}(\sin(x^3 + 5x)) = \cos(3x^2 + 5)$.

false

(c) If $f'(a) = 0$, and $f''(x) > 0$ then $f(a)$ is a local minimum.



cu

TRUE

(d) If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ then then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

False

3. Circle the best answer in the { brackets }.

(a) {3 points} If $f'(x) < 0$ for $-1 < x < 4$, then f is { increasing, decreasing, constant, concave down } on the interval $(-1, 4)$.

a function
discontinuous piecewise

(b) {6 points} If f is { increasing, decreasing, continuous, defined } on a closed interval $[a, b]$, then f attains a { absolute maximum, local maximum, inflection point } $f(c)$ at some number c in (a, b) .

(c) {3 points} If $f(c)$ is a local minimum, and $f'(c)$ exists, then $f'(c)$ is { an inflection point, 0, $\lim_{x \rightarrow a} f(x)$ }.

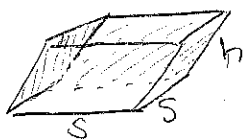
4. {5 points} Write an equation of the tangent to the curve of $f(x)$ at the point $(2, -7)$ if

$$f(x) = \frac{\sqrt{5-x^2}}{x+1}. \quad m = \frac{\sqrt{5-4}}{2+1} = \frac{1}{3}$$

$$y + 7 = \frac{1}{3}(x - 2)$$

5. {20 points} If 1200 cm^2 of material is available to make a box with a square base and an open top, follow the steps below to find the largest possible volume of the box.

- (a) Label the picture of the box below so that the square base has side length s and the height is h .



- (b) The surface area of this box is $A = s^2 + 4sh$. Solve this equation for h when $A = 1200$.

$$A - s^2 = 4sh$$

$$\frac{1200 - s^2}{4s} = h$$

- (c) The volume of the box is $V = s^2h$, substitute the h you found above into this equation.

$$V = s^2 \left(\frac{1200 - s^2}{4s} \right) = \frac{s(1200 - s^2)}{4} = \frac{1}{4}(1200s - s^3)$$

- (d) The derivative of $V' = \frac{1200 - 3s^2}{4}$. Solve for s when $V' = 0$.

$$0 = 1200 - 3s^2$$

$$3s^2 = 1200$$

$$s^2 = 400 \quad s = 20$$

- (e) The value of s you solved for in part (d) is a local maximum of the function describing the possible volumes of the box. It is the length of the sides of the base of the box that maximizes volume of the box. What should the height of the box be (ie what is h)?

$$1200 - 400 = h$$

$$800 = h$$

- (f) Using the dimensions you found in parts (d) and (e), what is the volume of the box?

$$800 \cdot 400 = 320000 \text{ cm}^3$$

6. {5 points each} Calculate y' .

(a) $y = (x^2 + 3x)^4$

$$y' = 4(x^2 + 3x)(2x + 3)$$

(b) $y = 3x^n + \cos(x)$

$$y' = 3nx^{n-1} - \sin x$$

(c) $y = \frac{x^2}{3x+1}$

$$y' = \frac{2x(3x+1) - x^2(3)}{(3x+1)^2}$$

(d) $y = \sin^{-1}(x^4)$

$$y' = \frac{4x^3}{\sqrt{1-x^8}}$$

(e) $2x = x^2 + y^2$

$$2 = 2x + 2yy'$$

$$\frac{2-2x}{2y} = y' = \frac{1-x}{y}$$

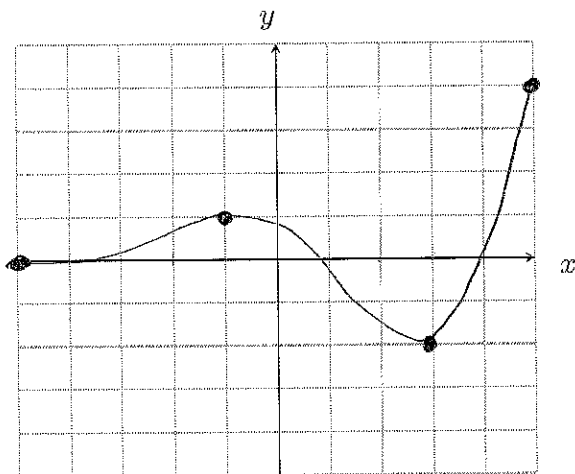
(f) $\sin(y) = ye^x$

$$y' \cos y = y'e^x + ye^x$$

$$y' = \frac{ye^x}{\cos y - e^x}$$

7. {12 points}

- (a) If possible, draw a continuous graph on the interval $[-5, 5]$ with the following characteristics: $f(-5) = 0$, $f(5) = 4$, $f(3) = -2$ is the only local minimum, and $f(-1) = 1$ is the only local maximum.



- (b) State the absolute maximum and absolute minimum values of the graph you drew above (if any).

$$\text{Abs min} = -2$$

$$\text{Abs max} = 4$$

{4 points} **Bonus Question:** This is for extra credit.

Use logarithmic differentiation to find y' if $y = x^{\cos(x)}$.

$$\ln y = \ln x^{\cos x}$$

$$\ln y = \cos x \ln x$$

$$\frac{y'}{y} = -\sin x \ln x + \frac{\cos x}{x}$$

$$y' = \frac{\cos x - \sin x \ln x}{x \cdot x^{\cos x}}$$